Lecture 3

Stochastic Analysis on Hilbert Spaces

1.3. Stochastic Integration on Hilbert spaces

1.3.2. Stochastic integral. Throughout this subsection let U and H be separable real Hilbert spaces and (W_t) be an U-valued Q-Wiener process w.r.t. $(\mathcal{F}_t)$. We want to construct the H-valued stochastic integral $\int_0^t \Phi(s)\, dW_s$ for appropriate integrands Φ in four steps similar to the real-valued case.

Step 1 Integration of elementary processes

$$\Phi(t) = \sum_{m=0}^{K-1} \Phi_m 1_{]t_m, t_{m+1}]}(t) \tag{1.24}$$

with

- $0 = t_0 < t_1 < \ldots < t_K = T$
- $\Phi_m : \Omega \longrightarrow L(U,H)$, $\mathcal{F}_{t_m}$-measurable and separable, where $L(U,H)$ denotes the (linear) space of all linear, continuous operators $U \longrightarrow H$.

We then define the stochastic integral

$$\int_0^t \Phi(s) dW_s := \sum_{m=0}^{K-1} \Phi_m (W_{t_{m+1}\wedge t} - W_{t_m \wedge t}), \quad 0 \le t \le T$$

which leads to a linear mapping

$$I : \mathcal{E} \longrightarrow \mathcal{M}_T^2$$

where $\mathcal{E}$ denotes the (linear) space of all elementary prozesses of type (1.24).

Step 2 Wiener-Itô isometry

We want to formulate the Wiener-Itô-isometry for H-valued stochastic integrals. To this end we first need to introduce a new class of linear operators:

Definition 1.21. Let $L \in L(U,H)$ be a continuous linear operator from U to H. L is called Hilbert-Schmidt operator, if

$$\|L\|^2_{L_2(U,H)} := \sum_{k=1}^{\infty} \|Le_k\|_H^2 < \infty \tag{1.25}$$

for one (and hence all) CONS (e_k) of U. Let $L_2(U,H)$ be the (linear) space of all Hilbert-Schmidt operators L. Then $L_2(U,H)$ is again a real Hilbert space w.r.t. the scalar product

$$\langle A, B\rangle_{L_2(U,H)} := \sum_{k=1}^{\infty} \langle Ae_k, Be_k\rangle_H$$

(which is independent of the chosen CONS (e_k) (!)). The associated norm $\|\cdot\|_{L_2(U,H)}$ (as given in (1.25)) is called Hilbert-Schmidt norm.

Remark (separability of $L_2(U,H)$) A striking feature of the space $L_2(U,H)$ of Hilbert-Schmidt operators is the fact that it is again separable. Indeed, the set of finite rank operators

$$L_n u := \sum_{k=1}^{n} \langle u, e_k\rangle_U y_k\,, \quad y_k \in H, n \geq 1$$

is a separable dense subset of $L_2(U,H)$, since for any $L \in L_2(U,H)$ and L_n of the above type with $y_k := Le_k$ we have that $L_n e_k = Le_k$ for $k = 1,\dots,n$, and thus

$$\|L - L_n\|^2_{L_2(U,H)} = \sum_{k=n+1}^{\infty} \|Le_k\|^2 \to 0\,, n \to \infty.$$

In contrast to this, the space $L(U,H)$ is not separable, even not for separable Hilbert spaces U and H, unless U and H are both finite-dimensional.

Back to the stochastic intergal $\int_0^t \Phi(s)dW_s$. The following Wiener-Itô isometry holds:

$$\mathbb{E}\left(\left\|\int_0^t \Phi(s)dW_s\right\|_H^2\right) = \mathbb{E}\left(\int_0^t \|\Phi(s)\circ\sqrt{Q}\|^2_{L_2(U,H)}ds\right), \quad t \in [0,T] \tag{1.26}$$

In particular,

$$\left\|\int_0^{\cdot} \Phi(s)dW_s\right\|^2_{\mathcal{M}_T^2} = \mathbb{E}\left(\int_0^T \|\Phi(s)\circ\sqrt{Q}\|^2_{L_2(U,H)}ds\right).$$

Proof of the Wiener-Itô isometry.

W.l.o.g. $t = T$. To simplify notations let $\Delta_m := W(t_{m+1}) - W(t_m)$. Then

$$\begin{aligned}\mathbb{E}\left(\left\|\int_0^T \Phi(s)dW(s)\right\|_H^2\right) &= \mathbb{E}\left(\|\sum_{m=0}^{k-1}\Phi_m\Delta_m\|_H^2\right)\\ &= \sum_{m,n=0}^{k-1}\mathbb{E}\left(\langle\Phi_m\Delta_m,\Phi_n\Delta_n\rangle_H\right)\end{aligned} \tag{1.27}$$

W.l.o.g. let (e_k), $k \geq 1$, be a CONS of U consisting of eigenvectors of Q with corresponding eigenvalues λ_k, $k \geq 1$, and let $\beta_k(t) := \langle W(t), e_k\rangle_U$, $k \geq 1$, be the corresponding independent 1d-Brownian motions. For any m,n we then have that

(1.28)

$$\begin{aligned}&\mathbb{E}\left(\langle\Phi_m\Delta_m,\Phi_n\Delta_n\rangle_H\right)\\ &\quad= \sum_{k,l=1}^{\infty}\sqrt{\lambda_k}\sqrt{\lambda_l}\mathbb{E}\Big(\langle\Phi_m(e_k),\Phi_n(e_l)\rangle_H(\beta_k(t_{m+1})-\beta_k(t_m))(\beta_l(t_{n+1})-\beta_l(t_n))\Big)\end{aligned}$$

To further simplify notations, let $\Delta_n\beta_l := \beta_l(t_{n+1}) - \beta_l(t_n)$.

If $m \neq n$ or $k \neq l$ then

$$\mathbb{E}(\langle\Phi_m(e_k),\Phi_n(e_l)\rangle_H\Delta_m\beta_k\Delta_n\beta_l) = 0$$

since the Brownian increments $\Delta_m\beta_k$ and $\Delta_n\beta_l$ are independent of $\mathcal{F}_{m\wedge n}$, so that in case of $n \neq m$ or $k \neq l$

$$\mathbb{E}(\Delta_m\beta_k\Delta_n\beta_l \mid \mathcal{F}_{m\wedge n}) = \mathbb{E}(\beta_k(t_{m+1}) - \beta_k(t_m) \mid \mathcal{F}_{m\wedge n})\mathbb{E}(\beta_l(t_{n+1}) - \beta_l(t_n) \mid \mathcal{F}_{m\wedge n}) = 0$$

and therefore in this case

(1.29)
$$\mathbb{E}(\Phi_m\Delta_m\beta_k\Phi_n\Delta_n\beta_l) = \mathbb{E}(\Phi_m\Phi_n\mathbb{E}((\beta_k(t_{m+1}) - \beta_k(t_m))(\beta_l(t_{n+1}) - \beta_l(t_n)) \mid \mathcal{F}_{m\wedge n})) = 0.$$

In the remaining case $m = n$ and $k = l$ we have that

$$\mathbb{E}(\langle\Phi_m(e_k), \Phi_n(e_l)\rangle_H\Delta_m\beta_k\Delta_n\beta_l) = \mathbb{E}\left(\|\Phi_m(e_k)\|_H^2\right)(t_{m+1} - t_m). \tag{1.30}$$

Inserting both identities (1.29) and (1.30) into (1.28) yields the assertion, since

$$\begin{aligned}
\mathbb{E}\left(\left\|\int_0^T \Phi(s)dW(s)\right\|_H^2\right) &= \sum_{m=0}^{K-1}\sum_{k=1}^{\infty}\lambda_k\mathbb{E}\left(\|\Phi_m(e_k)\|_H^2\right)(t_{m+1} - t_m)\\
&= \sum_{m=0}^{K-1}\sum_{k=1}^{\infty}\mathbb{E}\left(\|\Phi_m(\sqrt{\lambda_k}e_k)\|_H^2\right)(t_{m+1} - t_m)\\
&= \sum_{m=0}^{K-1}\sum_{k=1}^{\infty}\mathbb{E}\left(\|\Phi_m(\sqrt{Q}e_k)\|_H^2\right)(t_{m+1} - t_m)\\
&= \sum_{m=0}^{K-1}\sum_{k=1}^{\infty}\mathbb{E}\left(\|(\Phi_m\circ\sqrt{Q})(e_k)\|_H^2\right)(t_{m+1} - t_m)\\
&= \mathbb{E}\left(\sum_{m=0}^{K-1}\|\Phi_m\circ\sqrt{Q}\|_{L_2(U,H)}^2(t_{m+1} - t_m)\right)\\
&= \mathbb{E}\left(\int_0^T\|\Phi(s)\circ\sqrt{Q}\|_{L_2(U,H)}^2 ds\right).
\end{aligned}$$

Herein we used the fact that $Qe_k = \lambda_k e_k$ and therefore

$$\lambda_k\|\Phi_m e_k\|_H^2 = \left\|\Phi_m\left(\sqrt{\lambda_k}e_k\right)\right\|_H^2 = \left\|\Phi_m\circ\sqrt{Q}e_k\right\|_H^2.$$

□

The Wiener-Itô isometry now implies that

$$I : \mathcal{E} \longrightarrow \mathcal{M}_T^2$$

is the isometry

$$\|I(\Phi)\|_{\mathcal{M}_T^2}^2 = \|\Phi\|_T^2,$$

if we endow $\mathcal{E}$ with the seminorm

$$\|\Phi\|_T^2 := \mathbb{E}\left(\int_0^T\|\Phi\circ\sqrt{Q}\|_{L_2(U,H)}^2 ds\right).$$

Step 3 We can now extend I uniquely to a linear isometry

$$\overline{I} : \overline{\mathcal{E}} \longrightarrow \mathcal{M}_T^2,$$

where $\overline{\mathcal{E}}$ denotes the abstract completion of $\mathcal{E}$ w.r.t. the seminorm $\|\cdot\|_T$. The 3rd step now in the construction/definition of the stochastic integral consists in the alternative characterization of the abstract completion $\overline{\mathcal{E}}$ of admissible integrands. To this end we define

$$\begin{aligned}\mathcal{P}_T &:= \sigma\left(\{]s,t] \times F_s : 0 \leq s < t \leq T, F_s \in \mathcal{F}_s\} \cup \{\{0\} \times F_0 : F_0 \in \mathcal{F}_0\}\right)\\ &= \sigma\left(\{(H_s) : (H_s) \text{ left-cont. and } (\mathcal{F}_t) - \text{adapted}\}\right)\\ &= \text{predictable } \sigma - \text{algebra.}\end{aligned}$$

Remark Sometimes in the literature $\mathcal{P}_T$ is also called the previsible-σ-algebra.

With this definition we now have that

$$\overline{\mathcal{E}} = \text{Ł}^2(\Omega_T, \mathcal{P}_T, \mathbb{P}_T; L_2^0)/_\sim$$

with $\Omega_T = \Omega \times [0,T]$ and $\mathbb{P}_T = \mathbb{P} \otimes dt$. Here, L_2^0 denotes the space of all continuous linear operators $T : U \to H$ such that $T \circ \sqrt{Q}$ is Hilbert-Schmidt and "$\sim$" denotes the equivalence relation

$$\Phi \sim \Psi \text{ if } \Phi \circ \sqrt{Q} = \Psi \circ \sqrt{Q}\ \mathbb{P}_T - a.e.$$

Step 4 Localization

We use the same localization as in the finite-dimensional stochastic integration theory. This allows us to extend the construction/definition of the stochastic integral

$$\int_0^t \Phi(s) dW(s)\,, t \in [0,T],$$

to all

$$\Phi : \Omega_T \longrightarrow L(U,H)\,, \text{adapted, } \mathcal{P}_T - \text{measurable,}$$

satisfying

$$\mathbb{P}\left(\int_0^T \|\Phi(s) \circ \sqrt{Q}\|^2_{L_2(U,H)} ds < \infty\right) = 1.$$

Properties of the stochastic integral

(i) **Linearity** If $L \in L(H, \tilde{H})$ then

$$L\left(\int_0^t \Phi(s) dW(s)\right) = \int_0^t L \circ \Phi(s) dW(s).$$

(ii) Let $f : \Omega_T \longrightarrow H$, be $(\mathcal{F}_t)$-adapted and continuous. Then the real-valued stochastic integral

$$\int_0^t \langle f(s), \Phi(s) dW(s)\rangle_H := \int_0^t \tilde{\Phi}_f(s) dW(s)$$

with

$$\tilde{\Phi}_f(s)(u) := \langle f(s), \Phi(s)u\rangle_H, u \in U,$$

is well-defined.

(iii) Let $p \geq 1$. Then there exists a universal constant c_p such that

$$\mathbb{E}\left(\left\|\int_0^t \Phi(s)\,dW(s)\right\|_H^{2p}\right) \leq c_p \mathbb{E}\left(\int_0^t \|\Phi(s)\|_{L_2^0}^2\,ds\right)^p .$$

In particular, the following inequality, called the Burkholder-Davis-Gundy inequality, holds

$$\mathbb{E}\left(\sup_{t\in[0,T]}\left\|\int_0^t \Phi(s)\,dW(s)\right\|_H^{2p}\right) \leq c_p \mathbb{E}\left(\int_0^T \|\Phi(s)\|_{L_2^0}^2\,ds\right)^p$$

(iv) Quadratic variation: Let $M_t := \int_0^t \Phi(s)\,dW(s)$, then

$$\langle M\rangle_t := \int_0^t \|\Phi(s)\|_{L_2^0}^2\,ds\,, t \in [0,T]\,,$$

is the unique continuous increasing $(\mathcal{F}_t)_{t\geq 0}$-adapted process starting at zero such that

$$\|M_t\|_H^2 - \langle M\rangle_t\,, t \in [0,T]\,,$$

is an $(\mathcal{F}_t)_{t\in[0,T]}$-martingale. It can be shown that for any sequence of partitions $(\tau_n)_{n\geq 1}$ of $[0,T]$ with $\lim_{n\to\infty}|\tau_n| = 0$ it follows that

$$\lim_{n\to\infty}\sum_{t_i\in\tau_n, t_i\leq t}\|M_{t_{i+1}} - M_{t_i}\|_H^2 = \langle M\rangle_t \tag{1.31}$$

uniformly in t, in probability. More general, given any $h \in H$, the process

$$\int_0^t \|\sqrt{Q}\circ\Phi(s)^* h\|_U^2\,ds\,, t \in [0.T],$$

is the unique continuous $(\mathcal{F}_t)_{t\in[0,T]}$-adapted process starting at zero such that

$$\langle M_t, h\rangle_H^2 - \int_0^t \|\sqrt{Q}\circ\Phi(s)^* h\|_U^2\,ds\,, t \in [0,T]\,,$$

is an $(\mathcal{F}_t)_{t\in[0,T]}$-martingale. In analogy with (1.31)

$$\lim_{n\to\infty}\sum_{t_i\in\tau_n, t_i\leq t}\left(\langle M_{t_{i+1}}, h\rangle_H - \langle M_{t_i}, h\rangle_H\right)^2 = \int_0^t \|\sqrt{Q}\circ\Phi(s)^* h\|_U^2\,ds$$

uniformly in t, in probability.

(v) Regularity of the stochastic integral: let $\alpha < \frac{1}{2}$ be given. Then for any $\Phi \in \mathcal{L}^2(\Omega_T, \mathcal{P}_T, \mathbb{P}_T; L_2^0)$

$$M_t = \int_0^t \Phi(s)\,dW(s) \in W^{\alpha,2}([0,T];H)$$

where for a given Banach space E, the space $W^{\alpha,2}([0,T];E)$ consists of all functions $M \in L^2([0,T];E)$ satisfying

$$\int_0^T\int_0^T \frac{\|M_s - M_t\|_E^2}{|s-t|^{1+2\alpha}}\,ds\,dt < \infty\,.$$

Proof: The Wiener-Itô isometry implies that

$$\begin{aligned}\mathbb{E}\left(\int_0^T\int_0^T \frac{\|M_t-M_s\|_H^2}{|t-s|^{1+2\alpha}}\,ds\,dt\right) &= \int_0^T\int_0^T \frac{\mathbb{E}\left(\int_{s\wedge t}^{s\vee t}\|\Phi(r)\|_{L_2^0}^2\,dr\right)}{|t-s|^{1+2\alpha}}\,ds\,dt\\ &= 2\int_0^T\int_t^T \frac{\int_t^s \mathbb{E}\left(\|\Phi(r)\|_{L_2^0}^2\right)dr}{|t-s|^{1+2\alpha}}\,ds\,dt\\ &\le \ldots \le C(\alpha)\int_0^T \mathbb{E}\left(\|\Phi(r)\|_{L_2^0}^2\right)dr\,.\end{aligned}$$

1.3.3. Appendix: Stochastic integration w.r.t. cylindrical Wiener processes. The construction of stochastic integrals $\int_0^t \Phi(s)\,dW(s)$ can be extended to the case where the covariance operator Q is only bounded, but not necessarily of finite trace. To this end, one needs to extend the notion of a Q-Wiener process. To simplify the presentation, we restrict ourselves to the particular case $Q = I$.

The representation of the Q-Wiener process obtained in Proposition 1.13 leads in the case $Q = I$ to the infinite series

$$W(t) = \sum_{k=1}^{\infty} \beta_k(t) e_k\,, t \in [0,T]\,,$$

for independent one-dimensional Brownian motions β_k, $k \ge 1$. Note that this series does not converge in U, since

$$\left\|\sum_{k=1}^n \beta_k(t)e_k\right\|_U^2 = \sum_{k=1}^n \beta_k(t)^2$$

and thus

$$\mathbb{E}\left(\left\|\sum_{k=1}^n \beta_k(t)e_k\right\|_U^2\right) = \sum_{k=1}^n \mathbb{E}(\beta_k(t)^2) = n\cdot t \uparrow \infty$$

for $n \to \infty$. However, for any Hilbert space $(U_1, \langle\ ,\ \rangle_{U_1})$ for which there exists a Hilbert-Schmidt embedding $J : U \to U_1$ the infinite series converges in U_1, since

$$\begin{aligned}\mathbb{E}\left(\left\|J\left(\sum_{k=1}^n \beta_k(t)e_k\right)\right\|_{U_1}^2\right) &= \sum_{k,l=1}^n \mathbb{E}\left(\langle \beta_k(t)J(e_k), \beta_l(t)J(e_l)\rangle_{U_1}\right)\\ &= \sum_{k=1}^n \mathbb{E}\left(\beta_k(t)^2\right)\|J(e_k)\|_{U_1}^2 \text{ since } \mathbb{E}\left(\beta_k(t)\beta_l(t)\right) = \delta_{kl}t\\ &= \sum_{k=1}^n t\,\|J(e_k)\|_{U_1}^2 \uparrow t\,\|J\|_{L_2(U,U_1)}^2\,.\end{aligned}$$

Remark 1.22. U_1 with the above properties always exists. For example, choose a sequence $(\alpha_k)_{k\ge1} \in \ell^2$ with $\alpha_k \ne 0$ for all k, let $U_1 = U$ and define

$$J : U \to U_1\,, u \mapsto \sum_{k=1}^{\infty} \alpha_k \langle u, e_k\rangle_U e_k\,.$$

In the following we fix a sequence of independent one-dimensional Brownian motions β_k, $k \geq 1$, a CONS $(e_k)_{k\geq 1}$ of U and a Hilbert space U_1 for which there exists a Hilbert-Schmidt embedding $J : U \to U_1$. In particular, $Q_1 := J \circ J^* \in L(U_1)$ is symmetric, positive definite with finite trace and the infinite series

$$W_1(t) = \sum_{k=1}^{\infty} \beta_k(t) J(e_k)\,, t \in [0,T]\,,$$

converges in $\mathcal{M}_T^2(U_1)$ and defines a Q_1-Wiener process on U_1.

For a given predictable process Φ satisfying

$$\mathbb{P}\left(\int_0^T \|\Phi(s)\|^2_{L_2(U,H)}\, ds < \infty\right) = 1\,,$$

using

$$\|\Phi(s)\|^2_{L_2(U,H)} = \|\Phi(s) \circ J^{-1}\|^2_{L_2(\sqrt{Q_1}(U_1),H)}\,,$$

we conclude that the stochastic integral

$$\int_0^t \Phi(s) \circ J^{-1}\, dW_1(s)$$

w.r.t. the Q_1-Wiener process is well-defined. Finally, we set

$$\int_0^t \Phi(s)\, dW(s) := \int_0^t \Phi(s) \circ J^{-1}\, dW_1(s)\,.$$

The class of admissible integrands is given by

$$\begin{aligned}\mathcal{N}_W = \{\Phi : \Omega_T \to L_2(U,H) \mid \Phi \text{ predictable and} \\ \mathbb{P}\left(\int_0^T \|\Phi(s)\|^2_{L_2(U,H)}\, ds < \infty\right) = 1\}\,.\end{aligned}$$